

**TEST OF MATHEMATICS FOR
UNIVERSITY ADMISSION****TMUA/C1 P2****Challenge Paper****75 minutes****PAPER 2****INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but do not open this question paper until you are told that you may do so.

A separate answer sheet is provided. Please check you have one. You also require a soft pencil and an eraser.

This booklet contains Paper 2: Mathematical Reasoning, one of the two papers in the TMUA Challenge Paper.

It has twenty multiple-choice questions and lasts 75 minutes. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

Choose the one option you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working, but no extra paper is allowed. Only your responses on the answer sheet will be marked.

Dictionaries and calculators may NOT be used. Every answer is exact or a round number; if the arithmetic is turning ugly, the intended route has been missed.

Diagrams are not drawn to scale unless a question says otherwise.

Please wait to be told you may begin before turning this page.

This booklet consists of twenty questions, a marked answer key, and full solutions.

PAPER 2

Mathematical Reasoning



1. Which of the following is a counterexample to the claim that $n^2 + n + 41$ is prime for every positive integer n ?

- A) $n = 1$
- B) $n = 10$
- C) $n = 20$
- D) $n = 30$
- E) $n = 40$
- F) $n = 39$
- G) $n = 50$
- H) there is none

2. Consider the statement: if a and b are both rational then $a + b$ is rational. Which of the following is its contrapositive?

- A) If $a + b$ is rational then a and b are both rational.
- B) If a and b are not both rational then $a + b$ is irrational.
- C) If $a + b$ is irrational then at least one of a and b is irrational.
- D) If $a + b$ is irrational then neither a nor b is rational.
- E) If a or b is irrational then $a + b$ is irrational.
- F) If a and b are both irrational then $a + b$ is irrational.
- G) If $a + b$ is rational then at least one of a and b is rational.
- H) If at least one of a and b is irrational then $a + b$ is not rational.

3. For the quadratic $ax^2 + bx + c$ with $a \neq 0$ and a, b, c real, the condition $ac < 0$ is
- A) necessary but not sufficient for two distinct real roots.
 - B) necessary and sufficient for two distinct real roots.
 - C) neither necessary nor sufficient for two distinct real roots.
 - D) sufficient for two distinct real roots only when $b = 0$.
 - E) necessary but not sufficient for two distinct real roots of opposite sign.
 - F) sufficient but not necessary for two distinct real roots.
 - G) equivalent to the discriminant being positive.
 - H) necessary and sufficient for two distinct real roots of the same sign.

4. Consider these three statements about real numbers.

P : for every x there exists y such that $y^2 = x$.

Q : there exists x such that $xy = 0$ for every y .

R : for all x and y , if $xy = 0$ then $x = 0$ or $y = 0$.

Which are true?

A) none of them

B) P only

C) Q only

D) R only

E) P and Q

F) P and R

G) Q and R

H) P , Q and R

5. The numbers $1, 2, 3, \dots, 2026$ are written on a board. Repeatedly, two of the numbers, a and b , are rubbed out and $|a - b|$ is written in their place. After 2025 such steps a single number is left. Which of the following must be true of it?

- A) It is 0.
- B) It is 1.
- C) It is odd.
- D) It is even.
- E) It is a multiple of 1013.
- F) It is at least 1013.
- G) It is prime.
- H) None of these must be true.

6. The following argument is offered, where a and b are non-zero real numbers.

- (1) $a = b$
- (2) $a^2 = ab$
- (3) $a^2 - b^2 = ab - b^2$
- (4) $(a - b)(a + b) = b(a - b)$
- (5) $a + b = b$
- (6) $2b = b$, so $2 = 1$

At which step does the argument first fail?

- A) step (2)
- B) step (3)
- C) step (4)
- D) step (5)
- E) step (6)
- F) step (1)
- G) more than one step fails
- H) it does not fail

7. Which statement best justifies the claim that $n^3 - n$ is divisible by 6 for every integer n ?
- A) n^3 and n always have the same parity.
 - B) 6 divides n^3 for every integer n .
 - C) $n^3 - n = n(n^2 - 1)$, and $n^2 - 1$ is always even.
 - D) The result holds for $n = 1, 2, 3$ and therefore for all n .
 - E) $n^3 \equiv n \pmod{6}$ because 6 has no repeated prime factor.
 - F) $n^3 - n = (n - 1)n(n + 1)$, and a product of three integers is always divisible by 6.
 - G) $n^3 - n$ is even, and every even number is divisible by 6.
 - H) $n^3 - n = (n - 1)n(n + 1)$ is a product of three consecutive integers, so one of them is a multiple of 3 and at least one is even.

8. Consider the claim: if 3 divides $a^2 + b^2$ then 3 divides a and 3 divides b , where a and b are integers. Which of the following is correct?

- A) True: a square is congruent to 0 or 1 modulo 3, so a sum of two squares is divisible by 3 only when both squares are.
- B) True, but only when a and b are both positive.
- C) False, as $a = 1$, $b = 2$ shows.
- D) False, as $a = 3$, $b = 6$ shows.
- E) True, and it follows at once from the fact that 3 is prime, with no further argument needed.
- F) False, as $a = 1$, $b = 5$ shows.
- G) True, because $a^2 + b^2 = (a + b)^2 - 2ab$ and 3 must therefore divide $2ab$.
- H) False, as $a = 2$, $b = 4$ shows.

9. A sequence (a_n) and a number L are given. Consider the statement

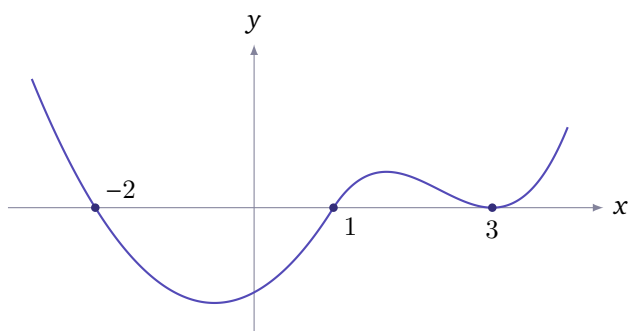
for every $\varepsilon > 0$ there exists N such that $|a_n - L| < \varepsilon$ for all $n > N$.

Which of the following is its negation?

- A) For every $\varepsilon > 0$ there exists N such that $|a_n - L| \geq \varepsilon$ for all $n > N$.
- B) There exists $\varepsilon > 0$ such that for every N there exists $n > N$ with $|a_n - L| \geq \varepsilon$.
- C) There exist $\varepsilon > 0$ and N such that $|a_n - L| \geq \varepsilon$ for all $n > N$.
- D) For every $\varepsilon > 0$ and every N there exists $n > N$ with $|a_n - L| \geq \varepsilon$.
- E) There exists $\varepsilon > 0$ such that $|a_n - L| \geq \varepsilon$ for every n .
- F) For every N there exist $\varepsilon > 0$ and $n > N$ with $|a_n - L| \geq \varepsilon$.
- G) There exists N such that for every $\varepsilon > 0$ there exists $n > N$ with $|a_n - L| \geq \varepsilon$.
- H) For every $\varepsilon > 0$ there exists N such that $|a_n - L| > \varepsilon$ for some $n > N$.

- 10.** Which of the following establishes that a 10×10 board cannot be tiled by 1×4 pieces, each of which covers four squares in a row or in a column?
- A)** 100 is not divisible by 4, so no tiling exists.
 - B)** Colour the board like a chessboard. Each piece covers two squares of each colour, and the two colours occur unequally often.
 - C)** No piece can be placed in a corner of the board.
 - D)** The board has an even number of squares, so any tiling would use an odd number of pieces.
 - E)** Nothing establishes it: a tiling does exist.
 - F)** Colour the square in row i and column j with $i + j$ modulo 4. Each piece covers all four colours once, so a tiling would need 25 squares of each colour; in fact they occur 25, 26, 25 and 24 times.
 - G)** Colour the columns in the repeating pattern 1, 2, 3, 4, 1, 2, ... Each piece covers all four colours, and the colours occur 30, 30, 20 and 20 times.
 - H)** A tiling would use 25 pieces, and 25 is odd.

11. The diagram shows a sketch of $y = f'(x)$, the derivative of a function f defined for all real x . The curve meets the x -axis at $x = -2$ and $x = 1$, and touches it at $x = 3$.



Which statement about f is true?

- A) f has a local minimum at $x = -2$ and a local maximum at $x = 1$.
- B) f has exactly two stationary points.
- C) f has a local maximum at $x = 1$ and a local minimum at $x = -2$.
- D) f has a local maximum at $x = -2$, a local minimum at $x = 1$, and a stationary point of inflection at $x = 3$.
- E) f is decreasing for $x > 3$.
- F) f has a local maximum at $x = 3$.
- G) f has a local minimum at $x = 3$.
- H) f is increasing for $-2 < x < 1$.

12. Five points are placed inside a square of side 2. Which argument shows that some two of them are a distance at most $\sqrt{2}$ apart?

- A) Divide the square into four squares of side 1; two of the points lie in the same one, and its diagonal is $\sqrt{2}$.
- B) The five points cannot all lie at corners of the square.
- C) The average of the ten distances between the points is less than $\sqrt{2}$.
- D) Divide the square into five regions; each point lies in one of them.
- E) No argument works: the four corners together with the centre is a counterexample.
- F) The square has diagonal $2\sqrt{2}$, and half of $2\sqrt{2}$ is $\sqrt{2}$.
- G) Divide the square into two rectangles; three of the points lie in the same one.
- H) Any five points in the plane include two at distance at most $\sqrt{2}$.

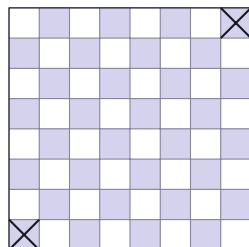
13. On an island there are 13 red, 15 green and 17 blue chameleons. Whenever two chameleons of different colours meet, both change to the third colour; nothing else changes a chameleon's colour. Which of the following states can be reached?

- A) All 45 red.
- B) All 45 green.
- C) All 45 blue.
- D) 15 of each colour.
- E) 22 red, 22 green and 1 blue.
- F) Any one of "all red", "all green" and "all blue".
- G) "All green", but neither of the other two single-colour states.
- H) None of the states listed above.

14. The inequality $x^2 + bx + c > 0$ holds for every real x if and only if

- A) $b^2 > 4c$
- B) $b^2 < 4c$
- C) $c > 0$
- D) $b^2 \leq 4c$
- E) $b^2 < 4c$ and $b > 0$
- F) $c > 0$ and $b > 0$
- G) $b^2 < 4c$ or $c > 0$
- H) $b^2 \geq 4c$

15. Two squares at opposite ends of a diagonal are removed from an 8×8 board, as shown. Can the remaining 62 squares be covered exactly by 31 dominoes, each covering two squares that share an edge?



- A) Yes, and an explicit covering can be written down.
- B) Yes, because 62 is even.
- C) No, because 62 is not divisible by 4.
- D) No, because no domino can be placed in a corner.
- E) Yes, because $62 = 2 \times 31$.
- F) No, because an odd number of rows is left uncovered.
- G) No: the two removed squares are the same colour, so 32 squares of one colour and 30 of the other remain, while every domino covers one of each.
- H) It depends on which pair of opposite corners is removed.

16. Let S be a set of ten distinct positive integers, each less than 100. Which of the following must be true?

- A)** S contains two elements whose sum is 100.
- B)** S contains three elements in arithmetic progression.
- C)** S contains two elements whose difference is 10.
- D)** The elements of S have sum at least 100.
- E)** S has two disjoint non-empty subsets with the same sum.
- F)** S contains two elements, one of which divides the other.
- G)** S contains a multiple of 10.
- H)** The largest element of S is at least 55.

17. Consider the statement for all real x , if $x > 1$ then $x^2 > x$. Which of the following is correct?
- A) The statement and its converse are both true.
 - B) The statement is false, as $x = \frac{1}{2}$ shows.
 - C) The converse is true and the contrapositive is false.
 - D) The statement and its contrapositive are true, but the converse is false, as $x = -1$ shows.
 - E) The statement, its converse and its contrapositive are all false.
 - F) The converse and the contrapositive are both true.
 - G) The statement is true and its contrapositive is false.
 - H) The statement is true, and its converse is false, as $x = 2$ shows.

18. How many of the ten digits can occur as the final digit of a perfect square?

- A) 4
- B) 5
- C) 6
- D) 7
- E) 8
- F) 9
- G) 10
- H) 3

19. For positive real a and b , which single fact is enough to prove that $\frac{a}{b} + \frac{b}{a} \geq 2$?

A) $(a - b)^2 \geq 0$

B) $(a + b)^2 \geq 0$

C) $a^2 + b^2 \geq 0$

D) $ab > 0$

E) $\frac{a}{b} > 0$ and $\frac{b}{a} > 0$

F) $a > 0$ and $b > 0$

G) $a^2 + b^2 > ab$

H) $(a - b)^2 > 0$

20. Consider the statement: for every integer n such that $n^2 + n$ is odd, n is a multiple of 7. Which of the following is correct?

- A)** It is false, as $n = 3$ shows.
- B)** It is true, vacuously: $n^2 + n = n(n + 1)$ is always even, so there is no such n .
- C)** It is false, as $n = 1$ shows.
- D)** It is true, and every such n happens to be a multiple of 7.
- E)** It is neither true nor false, because no such n exists.
- F)** It is false, as $n = 7$ shows.
- G)** It is true, but only because 7 is prime.
- H)** It cannot be decided without knowing which integers n are allowed.

Answer key

Paper 2			
Q	Ans	Q	Ans
1	E	11	D
2	C	12	A
3	F	13	H
4	G	14	B
5	C	15	G
6	D	16	E
7	H	17	D
8	A	18	C
9	B	19	A
10	F	20	B

Worked solutions

Paper 2: Mathematical Reasoning

1 (E). A counterexample must be an n for which the value is composite. The expression is Euler's, and it is prime for every n from 0 to 39, so all of $n = 1, 10, 20, 30, 39$ fail, giving the primes 43, 151, 461, 971 and 1601. At $n = 40$, though, $40^2 + 40 + 41 = 40(40 + 1) + 41 = 41 \times 41 = 1681$, a perfect square. The factorisation is visible without multiplying anything out, which is why 40 rather than some larger value is the natural place to look. Option G, $n = 50$, gives 2591, which is prime, so testing large values at random is a poor strategy here.

2 (C). The contrapositive of if P then Q is if not Q then not P , with both parts negated and the order reversed. Here P is a and b are both rational and Q is $a + b$ is rational. Negating P gives at least one of a and b is irrational, by De Morgan, not both are irrational. Option A is the converse, option E is the inverse, and options D, F and H all mis-negate the conjunction, which is the most common error.

3 (F). If $ac < 0$ then $-4ac > 0$, so $b^2 - 4ac \geq -4ac > 0$ and there are two distinct real roots: the condition is sufficient. It is not necessary, since $x^2 + 3x + 1$ has discriminant $5 > 0$ while $ac = 1 > 0$. Option G confuses a sufficient condition with an equivalent one; option E is wrong in the other direction, because $ac < 0$ is in fact both necessary and sufficient for roots of opposite sign, the product of the roots being $\frac{c}{a}$. Distinguishing sufficient from necessary is the whole question: one arrow, not two.

4 (G). P is false: taking $x = -1$, no real y has $y^2 = -1$. Q is true, witnessed by $x = 0$, which multiplies every y to zero. R is true: it is the statement that the reals have no zero divisors, and it follows from the fact that a non-zero real has a reciprocal, so $xy = 0$ with $x \neq 0$ forces $y = x^{-1} \cdot 0 = 0$. Hence Q and R only. Note how much work the order of the quantifiers does: reversing them in P would give a true statement about non-negative x , and reversing them in Q would give a false one.

5 (C). Adapted from British Mathematical Olympiad, Round 1. Look for a quantity the operation cannot change. Replacing a and b by $|a - b|$ changes the total on the board by $(a + b) - |a - b|$, which is either $2b$ or $2a$, and so is always even. The parity of the sum is therefore invariant. Initially the sum is

$$1 + 2 + \cdots + 2026 = \frac{2026 \times 2027}{2} = 1013 \times 2027,$$

a product of two odd numbers and hence odd. The last remaining number equals the sum at that moment, so it is odd. Nothing else is forced: the process can end at 1, and it can also end at large odd values, so options B, E, F and G all fail, while D contradicts the invariant.

6 (D). Every step up to (4) is sound: (4) is just (3) factorised. The failure is the passage from (4) to (5), which cancels the factor $a - b$ from both sides. Since step (1) asserts $a = b$, that factor is zero, and cancelling zero is not permitted. Steps (5) and (6) are then merely consequences of the illegitimate move, so the argument first fails at step (5). Naming step (6), where the absurdity becomes visible, confuses the symptom with the cause.

7 (H). Factorise: $n^3 - n = n(n^2 - 1) = (n - 1)n(n + 1)$, the product of three consecutive integers. Among any three consecutive integers exactly one is a multiple of 3, and at least one is even, so the product is divisible by 3 and by 2, and hence by 6 since 2 and 3 are coprime. Option F states the factorisation but then claims something false, that any product of three integers is divisible by 6; $1 \times 1 \times 1$ refutes it. Option C is a true factorisation attached to a false claim, since $n^2 - 1$ is odd when n is even. Option D mistakes checking three cases for a proof.

8 (A). Work modulo 3. Every integer is congruent to 0, 1 or -1 , so every square is congruent to 0 or 1. Hence $a^2 + b^2$ is congruent to 0, 1 or 2, and it is congruent to 0 only in the case $a^2 \equiv b^2 \equiv 0$, which forces $3 \mid a$ and $3 \mid b$. The claim is therefore true, and none of the offered counterexamples works: $1^2 + 2^2 = 5$, $1^2 + 5^2 = 26$ and $2^2 + 4^2 = 20$ are not multiples of 3 at all, while 3 and 6 satisfy the conclusion. Option E overreaches: primality alone does not give the result, as 2 shows, since $1^2 + 1^2 = 2$.

9 (B). Negate the quantifiers from the outside in, swapping each for every for there exists and vice versa, and negating the innermost statement last. For every ε becomes there exists ε ; there exists N becomes for every N ; for all $n > N$, $|a_n - L| < \varepsilon$ becomes there exists $n > N$ with $|a_n - L| \geq \varepsilon$. That is option B, and it says exactly what it should: some fixed error is exceeded infinitely often. Option C keeps the wrong quantifier on N and asserts far more, namely that the sequence stays away from L eventually; option D negates only part of the statement.

10 (F). Adapted from British Mathematical Olympiad, Round 1. Colour the square in row i and column j with the residue of $i + j$ modulo 4. Four squares in a row have consecutive values of $i + j$, and so do four in a column, so every piece covers each of the four colours exactly once. A tiling would use $\frac{100}{4} = 25$ pieces and hence 25 squares of each colour. Counting directly, the colours occur 25, 26, 25 and 24 times, so no tiling exists. Option A is false, since 4 does divide 100, and the chessboard colouring of option B is useless here, because a 10×10 board has 50 squares of each colour: this problem needs a colouring with four colours, not two. Option G uses four colours but the wrong ones, since a vertical piece then lies in a single column and covers one colour four times.

11 (D). The sketch is of f' , not f , so read signs rather than shape. $f' > 0$ for $x < -2$, $f' < 0$ for $-2 < x < 1$, and $f' > 0$ for $x > 1$ apart from the single point $x = 3$. So f rises, falls, then rises: a local maximum at $x = -2$ and a local minimum at $x = 1$. At $x = 3$ the derivative touches zero without changing sign, so f is stationary there but neither a maximum nor a minimum, a stationary point of inflection. That is three stationary points in all, which rules out B, and f is increasing, not decreasing, beyond $x = 3$, which rules out E. Options A and C swap the maximum and the minimum, the standard slip when the sign of f' is read backwards.

12 (A). Divide the square into four squares of side 1 by cutting along the two midlines. Five points in four regions force two points into the same small square, by the pigeonhole principle, and two points in a square of side 1 are at most its diagonal, $\sqrt{2}$, apart. Option E is tempting but wrong: the four corners together with the centre gives minimum distance exactly $\sqrt{2}$, satisfying the claim rather than refuting it, which is precisely why the bound is stated as at most. Option G divides into two 1×2 rectangles, whose diameter is $\sqrt{5}$, too weak to conclude anything.

13 (H). A classic olympiad invariant. When two chameleons of different colours meet, two of the three counts drop by 1 and the third rises by 2. Since $2 \equiv -1$, all three counts drop by 1 modulo 3, so every pairwise difference is unchanged modulo 3: two counts are congruent modulo 3 after a meeting exactly when two were congruent before it. Here 13, 15 and 17 leave remainders 1, 0 and 2, all different. Every state listed has two counts congruent modulo 3: in a monochromatic state the two zeros are congruent, 15, 15, 15 has three, and 22, 22, 1 has two. None of them is reachable, so the answer is H. Chasing individual sequences of meetings is hopeless; the invariant settles all five at once.

14 (B). $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$, whose least value is $c - \frac{b^2}{4}$, attained at $x = -\frac{b}{2}$. This is positive precisely when $b^2 < 4c$, so that condition is both necessary and sufficient. Option D allows equality, where the minimum is zero and the strict inequality fails at one point; option C is neither necessary nor sufficient, as $x^2 + 5x + 1$ shows; option E adds a condition that is not needed, as $x^2 - x + 3$ shows; option G weakens the conjunction to a disjunction, which the same example

refutes.

15 (G). A classic olympiad colouring. Every domino covers two squares sharing an edge, and adjacent squares of a chessboard have opposite colours, so each domino covers exactly one black and one white square. Thirty-one dominoes would therefore cover 31 of each. But the two removed squares lie at opposite ends of a diagonal and so have the same colour, leaving 32 squares of one colour and 30 of the other. The counts cannot match, so no covering exists. The argument is independent of which diagonal is used, since both pairs of opposite corners are monochromatic, which disposes of option H.

16 (E). Adapted from British Mathematical Olympiad, Round 1. Count subsets against sums. A ten-element set has $2^{10} - 1 = 1023$ non-empty subsets, while every subset sum lies between 1 and $90 + 91 + \dots + 99 = 945$. Since $1023 > 945$, two different subsets must have the same sum, by the pigeonhole principle; deleting their common elements leaves two disjoint subsets, still of equal sum, and neither is empty, since one subset containing the other and having the same sum would force the extra elements to sum to zero. Every other option fails for some admissible set: $\{1, 2, \dots, 10\}$ has no two elements summing to 100, no two differing by 10, total $55 < 100$, largest element 10, and no multiple of 10 if 10 is replaced by 11; $\{50, 51, \dots, 59\}$ has no element dividing another; and $\{1, 2, 4, 5, 10, 11, 13, 14, 28, 29\}$ contains no three terms in arithmetic progression.

17 (D). The statement is true: if $x > 1$ then multiplying the inequality $x > 1$ by the positive number x gives $x^2 > x$. Its contrapositive, if $x^2 \leq x$ then $x \leq 1$, is therefore also true, since a statement and its contrapositive are logically equivalent and need no separate proof. The converse, if $x^2 > x$ then $x > 1$, is false: $x = -1$ gives $1 > -1$ with $x < 1$. Option H offers $x = 2$, which satisfies both sides and so refutes nothing; option G forgets that the contrapositive cannot differ in truth value from the original.

18 (C). The final digit of n^2 depends only on the final digit of n , so ten cases settle it: 0, 1, 4, 9, 6, 5, 6, 9, 4, 1. The distinct values are 0, 1, 4, 5, 6, 9, so six digits occur, and 2, 3, 7, 8 never do. The symmetry of the list is not an accident, since n and $10 - n$ have squares agreeing modulo 10, so only six cases need checking in practice.

19 (A). Multiply through by ab , which is positive and so preserves the inequality: the claim becomes $a^2 + b^2 \geq 2ab$, that is $(a - b)^2 \geq 0$, which is true for all real a and b . Reversing those steps is a complete proof. Options D, E and F are needed to justify the multiplication but on their own prove nothing about the size of the two terms; option H is simply false when $a = b$, exactly the case where the original inequality is tight; and option G gives only $\frac{a}{b} + \frac{b}{a} > 1$, which is too weak.

20 (B). $n^2 + n = n(n + 1)$ is a product of consecutive integers, one of which is even, so it is even for every integer n and never odd. The hypothesis therefore holds for no n at all, and a conditional statement whose hypothesis is never satisfied is true by convention: there is no case in which it could fail. Option E misreads vacuous truth as a third truth value, and options A, C and F offer values of n that do not satisfy the hypothesis, so they cannot be counterexamples. The conclusion, being about multiples of 7, is a red herring: any conclusion at all would have done.

